Installing and Updating Firmware

Instructions to update the main firmware on Duet 3 MB6HC and Duet 3 Mini 5+ in standalone mode, Duet 2 WiFi, Ethernet and Maestro, Duet Web Control (DWC) and the WiFi firmware on Duet 3 Mini 5+ WiFi and Duet 2 WiFi boards.

Notes before you start

For updating the firmware on a Duet board in SBC mode, see SBC setup for Duet.
The instructions below are for updating the firmware on Duet 3 (all boards, in standalone mode), Duet 2 WiFi, Ethernet and Maestro boards only. Using them to update a Duet board using a Single Board Computer (eg Raspberry Pi) may leave your board in an unresponsive state.

- Check that you have downloaded the correct firmware before installing it. See section below regarding which files are needed for each board.
- Make sure you download and save the files as bin or zip and check the files sizes match.
- Backup your SD card and download a copy of your current firmware version in case you need to revert the firmware.
- Some updates to the main firmware on a Duet 2 WiFi or Duet 3 Mini 5+ WiFi may require an update to a compatible WiFi server firmware version as well, and vice versa. See the 'updating WiFi firmware' below which explains how to do this.
- **If you download more than just the main firmware (DuetWiFiFirmware.bin/Duet2CombinedFirmware.bin) then please read the 'updating more than one firmware at a time' section below before proceeding as the firmware may need to be updated in a specific order.**
- For RepRapFirmware 3, please ensure you read the RepRapFirmware 3.x Changelog notices for the new version of the firmware as it will contain information on changes that affect your configuration.
- For RepRapFirmware 2, please ensure you read the RepRapFirmware 1.x and 2.x Changelog notices for the new version of the firmware as it will contain information on changes that affect your configuration.

Duet boards and supported firmware versions

Duet 3 MB6HC - RepRapFirmware 3.x
Duet 3 MB6XD - RepRapFirmware 3.4 and later
Duet 3 Mini 5+) - RepRapFirmware 3.2 and later
Duet 2 (WiFi, Ethernet, Maestro) - RepRapFirmware 2.x, 3.x
Legacy Duet 0.6 and 0.85 - RepRapFirmware 1.x

Downloading RepRapFirmware files

The latest release, along with beta, release candidates (RC) and previous versions of RepRapFirmware are available at https://github.com/Duet3D/RepRapFirmware/releases

RepRapFirmware 2.x and 1.x are no longer developed. The last versions (dated 02/2020) are RepRapFirmware 2.05.1 and RepRapFirwmare 1.26.1, and are available here .

Our general advice is to use the 'Latest release' version from GitHub , unless you have a specific reason to use a 'beta' or 'RC' release. Beta and RC versions are marked 'Pre-release' on GitHub.

Each 'release' version has a large zip file that contains all the files needed to update all Duet boards, e.g. in RRF 3.3 it is 'Duet2and3Firmware-3.3.zip'. Download this, without unzipping it. Individual binary files can also be downloaded, as needed.

'Beta' and 'RC' releases are supplied as individual files, not as a convenient zip file, and need to be downloaded and installed individually.

Required files

Each Duet board needs up to four files to fully update the firmware and software. These are:

Firmware binary - This is the main firmware for the Duet boards, and there is a specific version for each Duet board. The main firmware for all boards except Duet 3 Mini 5+ is supplied as a .bin file; for the Duet 3 Mini 5+ it is a .uf2 file.

In-App programmer (IAP) binary - This file is needed on the SD card and does the job of programming the firmware. There is a specific version for each Duet board. They don't change with every firmware release, but new versions often accompany major firmware releases. You will get a message if it is missing, with the filename it expects, and (in DWC 3.3 and later) the location it expects to find it. If it is missing, check the firmware version your Duet is currently using, and download the iap file from the matching firmware version on the Duet3D Github repository . If the IAP file isn't included in the 'Assets' list of that firmware release, it should be in the .zip file for that release. Don't rename the iap file you have to the one it wants; most likely it won't work, but could cause other issues.

WiFi firmware binary - Required for the ESP8266-based WiFi board on the Duet 2 WiFi and Duet 3 Mini 5+ WiFi. Use the same version that is supplied with the firmware you are flashing.

Duet Web Control (DWC) zip - Web-browser-based user interface for communicating and controlling all Duet boards. Use the same version that is supplied with the firmware you are flashing.

Firmware file naming

Below is the firmware files, and file names, required for each Duet board. Filenames are correct for RRF 3.3 and 3.4. Older versions of the firmware will have similar file names, but they have changed over time. Note that these files are for Standalone mode, NOT SBC mode (except for Duet 2 WiFi/Ethernet in SBC mode). See SBC setup for Duet 3 for updating a Duet with connected SBC.

	Firmware binary	IAP binary	WiFi firmware binary	Duet Cont
Firmware .zip	Duet2and3Firmware-[X.X.X].zip	Large .zip file that contains all the files needed to update a boards. [X.X.X] is the firmware version, e.g. '3.4.0'. Upload direct to DWC, which will use and keep only the files it needs.		
Duet 3 Mainboard 6HC	Duet3Firmware_MB6HC.bin	Duet3_SDiap32_MB6HC.bin		Duet SD.zi
Duet 3 Mainboard 6XD	Duet3Firmware_MB6XD.bin	Duet3_SDiap32_MB6XD.bin		Duet SD.zi
Duet 3 Mini 5+	Duet3Firmware_Mini5plus.uf2	Duet3_SDiap32_Mini5plus.bin	DuetWiFiServer.bin	Duet SD.zi
Duet 3 Expansion 1HCL	Duet3Firmware_EXP1HCL.bin			
Duet 3 Expansion 1XD	Duet3Firmware_EXP1XD.bin			
Duet 3 Expansion 3HC	Duet3Firmware_EXP3HC.bin			
Duet 3 Toolboard 1LC	Duet3Firmware_TOOL1LC.bin			
SAMMY C21	Duet3Firmware_SAMMYC21.bin			
Duet 2 WiFi/Ethernet	Duet2CombinedFirmware.bin	Duet2_SDiap32_WiFiEth.bin	DuetWiFiServer.bin	Duet SD.zi
Duet 2 WiFi/Ethernet - SBC	Duet2Firmware_SBC.bin			

	Firmware binary	IAP binary	WiFi firmware binary	Duet Cont
Duet 2 Maestro	DuetMaestroFirmware.bin	Duet2_SDiap32_Maestro.bin		Duet SD.zi

When using a Duet 3 mainboard as an expansion board, the /firmware folder on the SD card in the mainboard-as-expansion-board needs to contain the CAN IAP file. Otherwise it won't be possible to update the expansion board over CAN.

	IAP binary
Duet 3 Mainboard 6HC	Duet3_CANiap32_MB6HC.bin
Duet 3 Mainboard 6XD	Duet3_CANiap32_MB6XD.bin
Duet 3 Mini 5+	Duet3_CANiap32_Mini5plus.bin

Updating from older firmware versions

If you are updating from an old version of the firmware, there may be additional steps you need to take. Send M115 to your Duet via USB or in the DWC console to check the current firmware version.

Updating without firmware erase

We generally recommend that firmware updates should be done in incremental steps, especially if you are updating from a very old version. We usually suggest updating in the following order:
RRF v1.x to **RRF v2.05.1** to **RRF v3.0** to **RRF v3.3** to **the latest stable release**.
You don't need to downgrade your firmware; just update to the next version on from your current version, eg if you are on RRF 3.1.1, update to v3.3, then the latest stable release.
See the tabs below for notes on updating to specific releases.

Updating from RRF 3.3 | Updating from RRF 3.0 - 3.2.2 | Updating from RRF 1.x or 2.x

If you are updating from RRF 3.3 or later, you should be able to update by downloading the Duet2and3Firmware-3.x.zip file (where 3.x is the version number) from the latest stable release. This contains all the firmware and software files for all Duet boards. There's no need to unzip it. You can download individual files as necessary, too. See the section later about updating individual files. Firmware files will be stored in the '0:/firmware/' directory.

Download file Duet2and3Firmware-X.X.zip file from the latest stable release, which is available from the Duet3D Github repository here . Follow the 'Usual procedure' instructions below to update the firmware.

Updating with firmware erase

While it is possible to update from older versions of RepRapFirmware by uploading firmware releases via the web interface, you may have to do this multiple times depending on the firmware version. If you're planning on jumping to the latest RRF 3.x version, it may be more convenient to simply erase the firmware and flash the latest version with BOSSA or SAM-BA; see 'Fallback procedure #2' instructions below.

File locations on the Duet SD card

- With all firmware versions up to RRF v3.2.2, all firmware update files are stored in the '0:/sys/' directory.
- From RRF v3.3, to avoid too many files in this folder, all firmware update files are stored in '0:/firmware/' directory.

This means if you have upgraded from, for example, RRF v3.2.2 to RRF v3.3beta1 (the firmware, IAP and DWC files will be put in the 'sys' directory), then update to RRF v3.3beta2 by just uploading the firmware binary (which it will put in the 'firmware' directory), you'll get a message that the IAP file is missing, because it expects to find it in 'firmware'. Upload the IAP file from the v3.2.2 release, and it should put it in the 'firmware' folder.

Should you need to recreate the SD card see the SD card page for a full description of the structure and contents, .

Usual procedure

For updating the firmware on a Duet board in SBC mode, see SBC setup for Duet.
The instructions below are for updating the firmware on Duet boards, in standalone mode. Using them to update a Duet board using a Single Board Computer (eg Raspberry Pi) may leave your board in an unresponsive state.

1. Choose which version of RRF you want to use to update your Duet. Download the firmware zip bundle (named Duet2and3Firmware-X.X.zip, where X.X is the firmware version number), or individual binary files, from the Duet3D Github repository .
   For the latest stable release, use https://github.com/Duet3D/RepRapFirmware/releases/latest
2. Connect to DuetWebControl.
3. In DWC, go to the System page and click the "Upload System Files" button.
4. Browse to the new firmware zip file or individual firmware binary file you have downloaded.
5. After uploading, Duet Web Control will ask you whether you want to install the new firmware. If you are updating using the firmware zip bundle, this will update firmware, IAP, WiFi firmware and DWC all at once, so

click **Yes**.

6. If you are only updating the main firmware, answer **Yes**. If you are updating WiFi firmware as well, answer **No** so that you can upload the new WiFi firmware too.
7. If your machine is idle when you upload the file but Duet Web Control does not offer to install it, then it doesn't recognise the filename as being a firmware binary. Check that you have uploaded the correct file.

Fallback procedure #1

If you are unable to connect to your Duet from a browser, but you can connect to the Duet via USB using YAT or other serial terminal:

Use caution when connecting your Duet to a PC via USB while VIN is still connected as this may lead to a ground loop situation. When connecting USB, and VIN power is not required, disconnect the PSU that supplies VIN power from the mains. See here for more details: USB ground loops

1. Take the on-board SD card out and put it in a computer
2. Delete any existing firmware files in /sys
3. Copy the new firmware binary file to the **/firmware** (if RRF 3.3beta1 or later is installed) or **/sys** folder (for earlier firmware versions)
4. Also copy the relevant IAP file (see Required Files, above) into **/sys**
5. If your board is a Duet WiFi, also copy the new DuetWiFiServer.bin file into /sys
6. Eject/safely remove the SD card from the PC
7. Replace the on-board SD card in the Duet 2 WiFi
8. Restart the Duet 2 WiFi and connect with Pronterface or YAT
9. Send command **M997 S0** to install the firmware.
10. Press the Disconnect button in Pronterface, then when the USB port disappears and reappears about 20 seconds later, press Connect.
11. Send M115 to confirm the new firmware version
12. If the board is a Duet WiFi, send **M997 S1** to install the WiFi module firmware

Note: when you run M997 S0, if your board is a Duet WiFi or Ethernet running old firmware, you may get an error message telling you that file DuetWiFiFirmware.bin or DuetEthernetFirmware.bin was not found. If that happens, copy or rename the Duet2CombinedFirmware.bin file to that filename and try again.

Duet 3 expansion board firmware can be manually updated with M997 Bn, where n is the CAN bus address of the expansion board.

Fallback procedure #2

Only use this if the firmware on the Duet has been corrupted, or if you have accidentally pressed the Erase button, so you can't connect over USB with YAT..

Use caution when connecting your Duet to a PC via USB while VIN is still connected as this may lead to a ground loop situation. When connecting USB, and VIN power is not required, disconnect the PSU that supplies VIN power from the mains. See here for more details: USB ground loops

Before proceeding, connect the Duet to your PC using a USB cable and check whether the Duet is visible in the port list. Under Windows you can do this by going to **Device Manager** and expanding **Ports (COM and LPT)**. If **Duet 3D printer electronics** is listed, you should be able to connect using Pronterface or YAT unless some other program is already using the port. If **Bossa Port** is listed then the firmware has already been erased. If neither is listed:

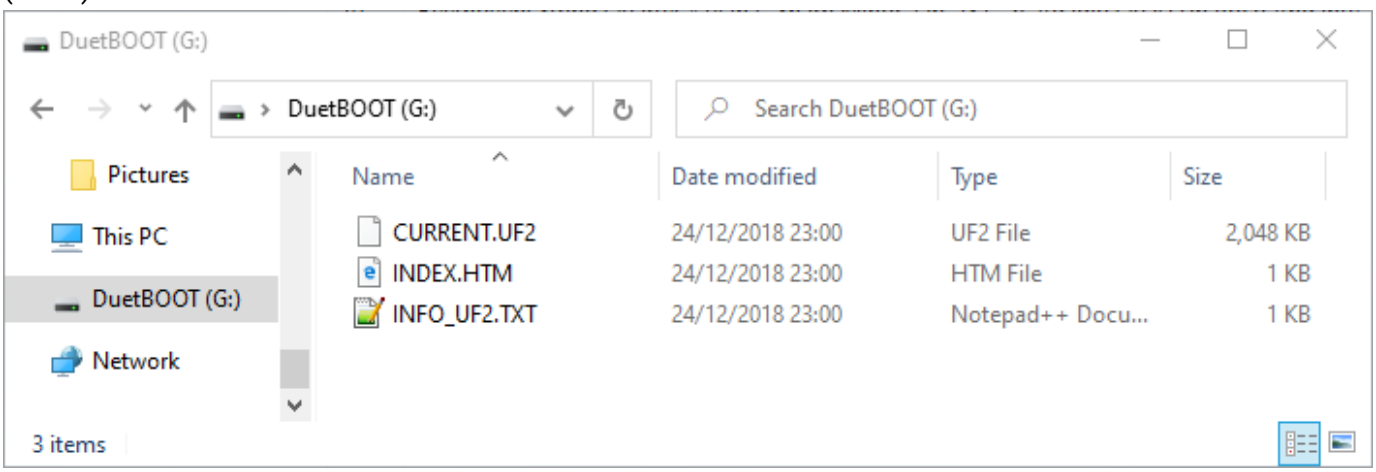
- Try a different USB cable
- Try a different USB port

If you get the **Duet 3D printer electronics** port to appear, then firmware is already running on the Duet and you should use fallback procedure #1 or #2 instead. If you get **Bossa Port**, read on.

Duet 3 Mini 5+ WiFi/Ethernet

The Duet 3 Mini 5+ supports an easy mechanism for updating the firmware directly.

- Connect the Duet to a PC (Windows, Mac or Linux) using a USB cable
- Press the Reset button twice quickly in succession. The Status LED should cycle slowly between dim and bright. This indicates that the bootloader is active.
- A new Mass Storage device should appear on the PC. Identify its drive letter (Windows) or mount path (Linux).



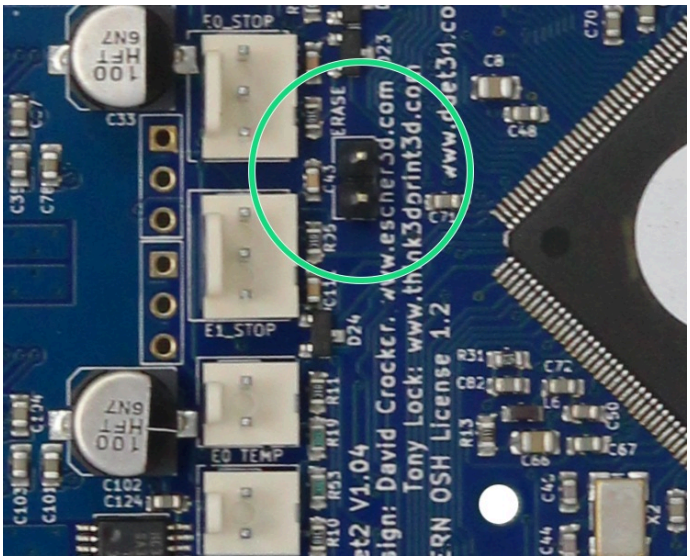
- A directory listing of this mass storage device should show three virtual files. File CURRENT.UF2 is the existing firmware on the Duet.
- If you don't want to update the firmware at this time, you can press reset once or cycle power to reboot the Duet with the existing firmware.

- Copy the new Duet3Firmware-Mini5plus.uf2 file to the mass storage device
- The Status LED will flicker as the firmware is being written, then the Duet will reboot using the new firmware. The mass storage device will disappear from your PC and the Duet 3 COM port will appear.

All other Duet boards

You will need a Windows, Apple Macintosh or Linux PC (can be Raspberry Pi).

- Download the main firmware file from



  - Pressing the **Erase** button (next to the reset button on older Duets); or
  - Jumpering the erase jumper (green circle in image on right)
- Remove the erase jumper after a few seconds.
- Then press the **Reset** button (orange circle in image on right).
- Check the DIAG LED, which will indicate if the Duet is in programming mode. See the wiring schematic of your board to find the location of this LED, though it is usually near the reset button.
  - On the MB6HC board the DIAG LED will either light up very dimly or not light up at all.
  - On the MB6XD board it will not light up at all.
  - On Duet 2, the Diag LED (between the USB connector and the SD card socket) will light up.
- Now use either Bossa or SAM-BA to flash the firmware to the Duet (see below).
- Duet 3 MB6XD pre-production (green) boards only: if you moved the Driver Enable Polarity Select jumper, move it back to the original position.

Bossa | SAM-BA

Bossa is available for Windows, Apple macOS and Linux.

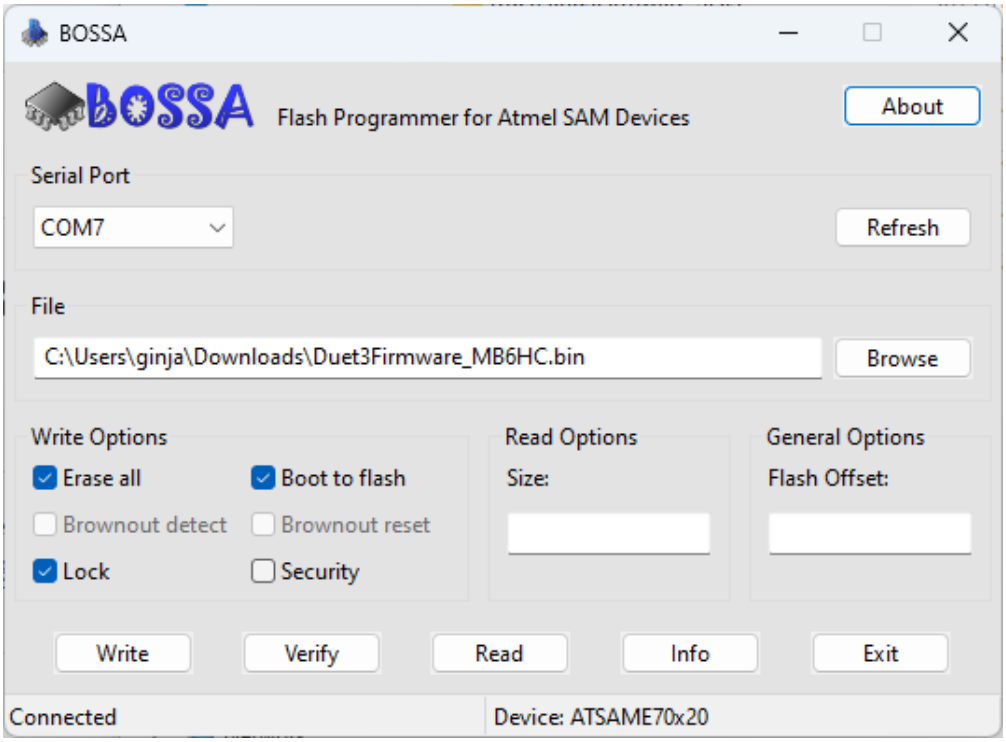
Notes

- **Duet 3 MB6HC and MB6XD:** When BOSSA is used to flash the SAME70 processor in the MB6HC, the processor uses the USB port in high speed mode (unlike the processors used in other Duets). This means that the length and quality of the USB cable is especially important. USB cables that can be used to send GCode commands and receive replies to/from a Duet MB6HC/MB6XD that is running RepRapFirmware may not be suitable for flashing the firmware using BOSSA.
- **Duet 3 MB6XD:** You must use the Duet3D version of BOSSA. If the Bossa port does not appear on the PC, check that the Driver Enable Polarity jumper is not in the wrong position (see earlier).
- If you have problems getting Bossa to connect to the Duet and you have an attached PaneDue, switch the PanelDue to the Setup page to stop it sending data to the Duet.

Windows | macOS | Raspberry Pi running DuetPi (after March 2022) | Linux - Debian/Ubuntu/Raspberry Pi OS | Compiling Bossa

- Download one of:
  - Duet3D version of Bossa (Windows only .exe file; supports Duet 2 and Duet 3 6HC/6XD)
  - ShumaTech Bossa v1.9.1 (Windows/Mac installer, supports Duet 2, does not support Duet 3 6HC/6XD).

- Open Bossa. It should detect the correct COM port automatically, if not press 'Refresh' and select the port from the drop down menu.
- Browse to the firmware binary file to be installed, and select it.
- **IMPORTANT:** Check boxes *Erase all*, *Lock*, and *Boot to flash*, then press *Write*.



- When the write completes, press *Verify* to confirm, then when complete, the reset button on the Duet.
- It is also possible to use bossac command line tool. Replace COMxx with the correct COM port number, and [path to file] with the path and name of the firmware binary, then send:
  `bossac -b -U -e -w -v -R --port=COMxx [path to file]`

Updating individual firmware components

Updating board firmware

- Upload the firmware binary file via DWC. You will be prompted if you want to install it, and the Duet will reboot after installation.

Updating WiFi firmware

Note, this applies to Duet 2 WiFi and Duet 3 Mini 5+ WiFi only.

- Check the current version of the WiFi firmware by sending M122, and checking the WiFi section at the end for the WiFi firmware version.

- Upload the new DuetWiFiServer.bin file to the /sys directory (RRF 3.2) or /firmware directory (RRF3.2.2 or later) on the on-board SD card, by either:
  - If WiFi is working, via the System page (click the "Upload System Files" button) of DuetWebControl
  - or by temporarily moving the on-board SD card to a PC. If you copy it directly to the SD card on a PC, you must remove the version number from the filename so that its name on the SD card is exactly DuetWiFiServer.bin. If you upload it through the Settings page of DuetWebControl, the renaming will be done automatically.
- If DWC doesn't ask if you want to install it, send command M997 S1 to install the new firmware. If you send this command from YAT or PanelDue, the update status will be reported.
- If you sent the M997 S1 command via the web interface, it will lose the connection and you should allow a minute or so for the firmware installation to complete before you press the Connect button.

Updating DuetWebControl files

- Upload the new **DuetWebControl.zip** file by going to the DWC System page and clicking the "Upload System Files" button. Select the file and DWC will extract the files, put them in the right place, and refresh the browser to start using the new files immediately.
- If the web interface is not working, then move the SD card to a PC, delete or rename any existing /www folder, create a new one, and extract the contents of DuetWebControl.zip into it. Do not unzip the individual .gz files, leave them as they are on the SD card.

Updating more than one firmware at a time

If you want to update the Duet firmware and WiFi firmware at the same time:

- Upload both firmware files via DWC System page (it will put them in the right place) but decline any offers to install them
- If you're editing the SD card on a PC, upload the files to /firmware or /sys (depending on firmware version) on the on-board SD card, then replace the card in the Duet.
- The M997 command will install firmware files, and can be run from DWC or via a serial terminal (eg YAT) connected by USB. Specifying which firmwares you want to update in the S parameter. For example, M997 S0:1 will update the mainboard and WiFi firmwares, provided that the firmware files and IAP binary are present in /firmware or /sys (depending on firmware version) on the SD card.

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